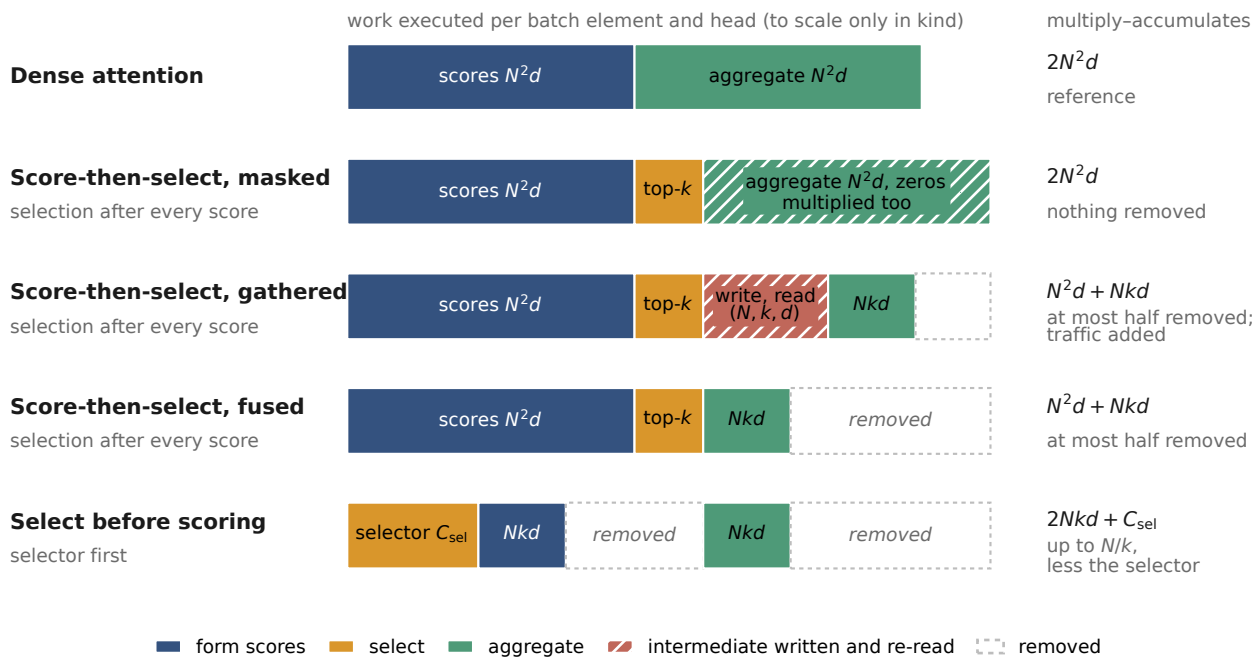
work executed per batch element and head (to scale only in kind)
multiply–accumulates
Dense attention
scores N^2d
aggregate N^2d
$2N^2d$
reference
Score-then-select, masked
selection after every score
scores N^2d
top-k
aggregate N^2d, zeros multiplied too
$2N^2d$
nothing removed
Score-then-select, gathered
selection after every score
scores N^2d
top-k
write, read (N, k, d)
Nkd
$N^2d + Nkd$
at most half removed; traffic added
Score-then-select, fused
selection after every score
scores N^2d
top-k
Nkd
removed
$N^2d + Nkd$
at most half removed
Select before scoring
selector first
selector C_{sel}
Nkd
removed
Nkd
removed
$2Nkd + C_{\text{sel}}$
up to N/k, less the selector
form scores
select
aggregate
intermediate written and re-read
removed